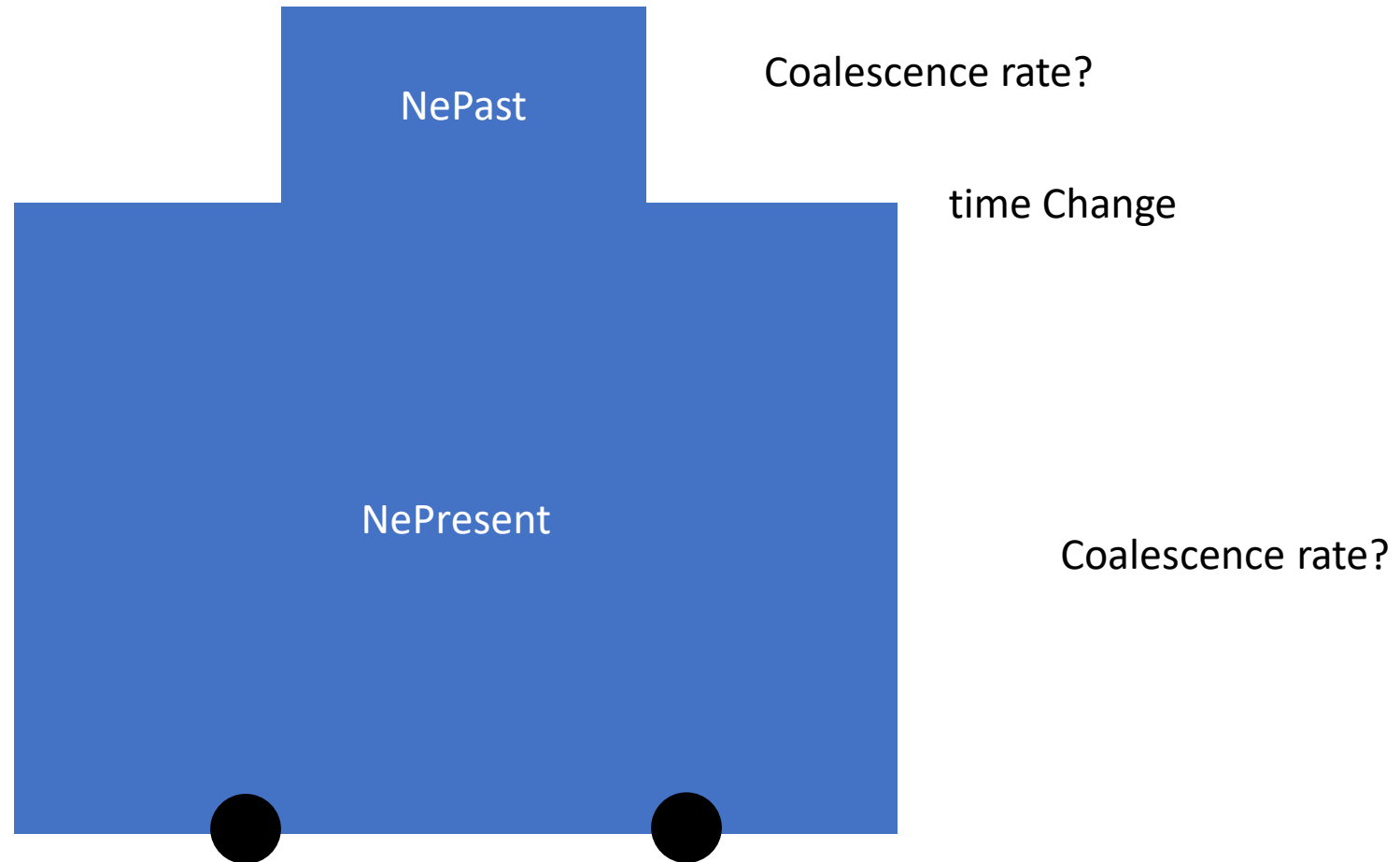


Practical session.

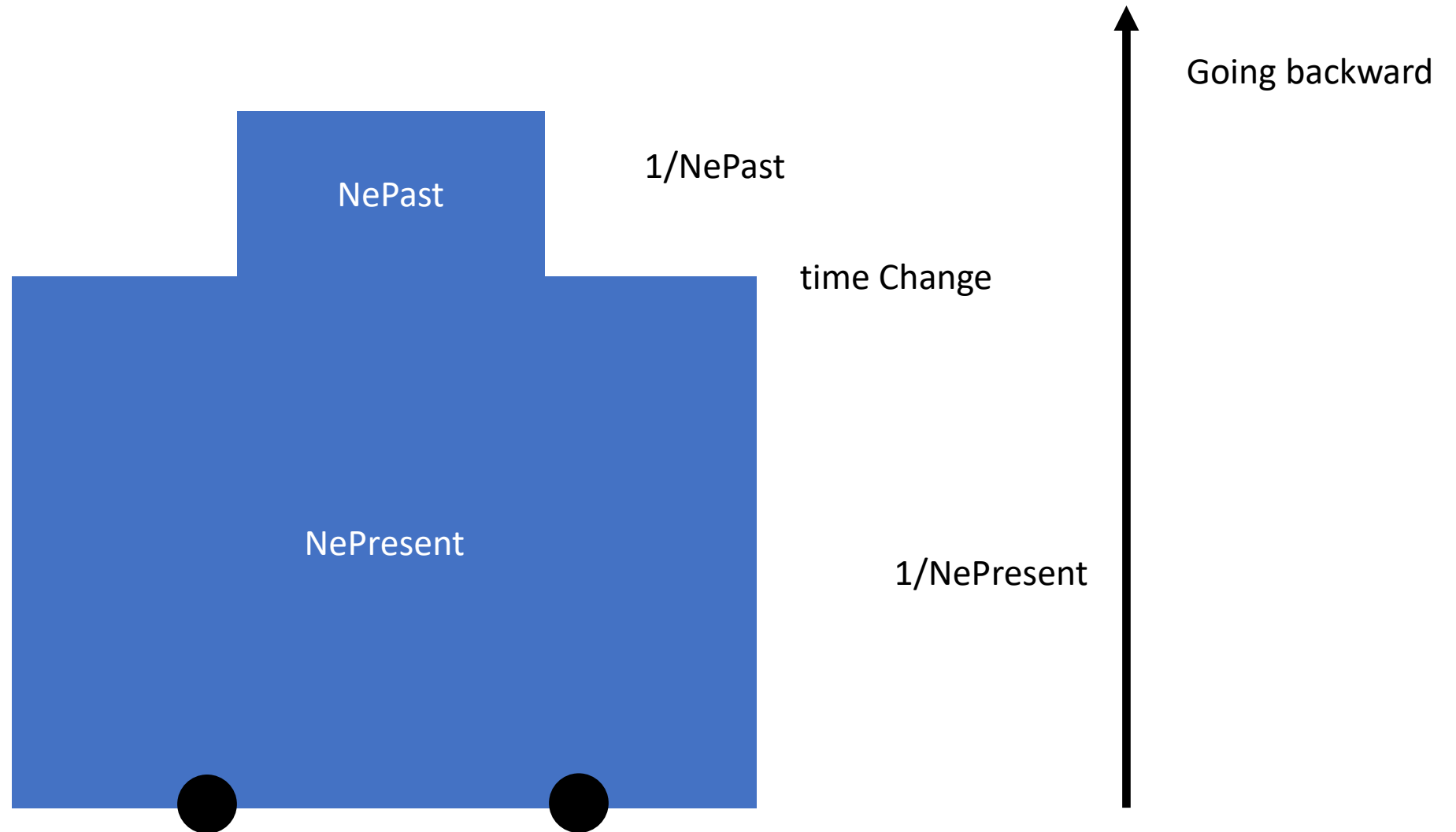
Coalescence and Relate

Oscar Lao

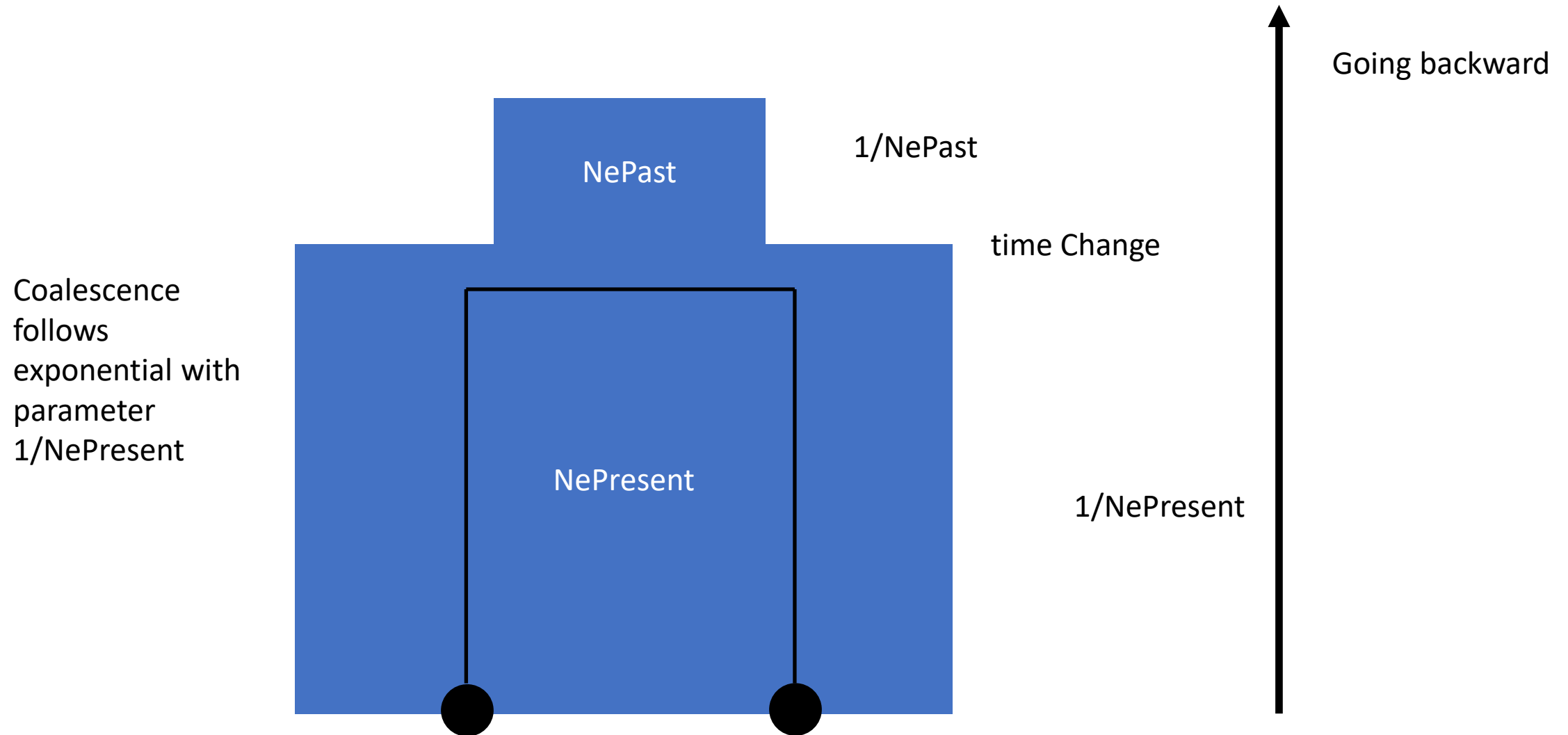
Model Ne changes



Model Ne changes

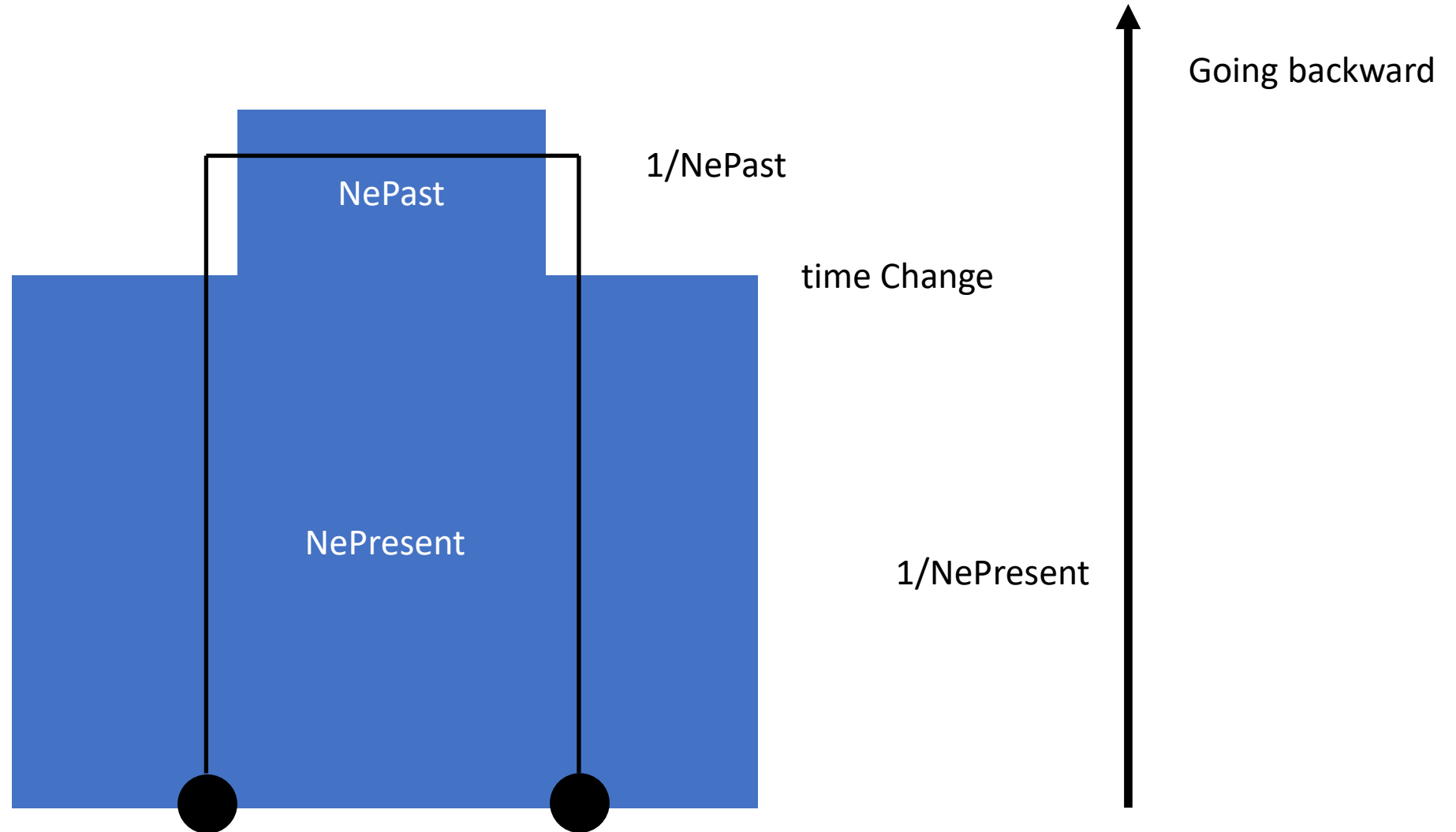


Model Ne changes



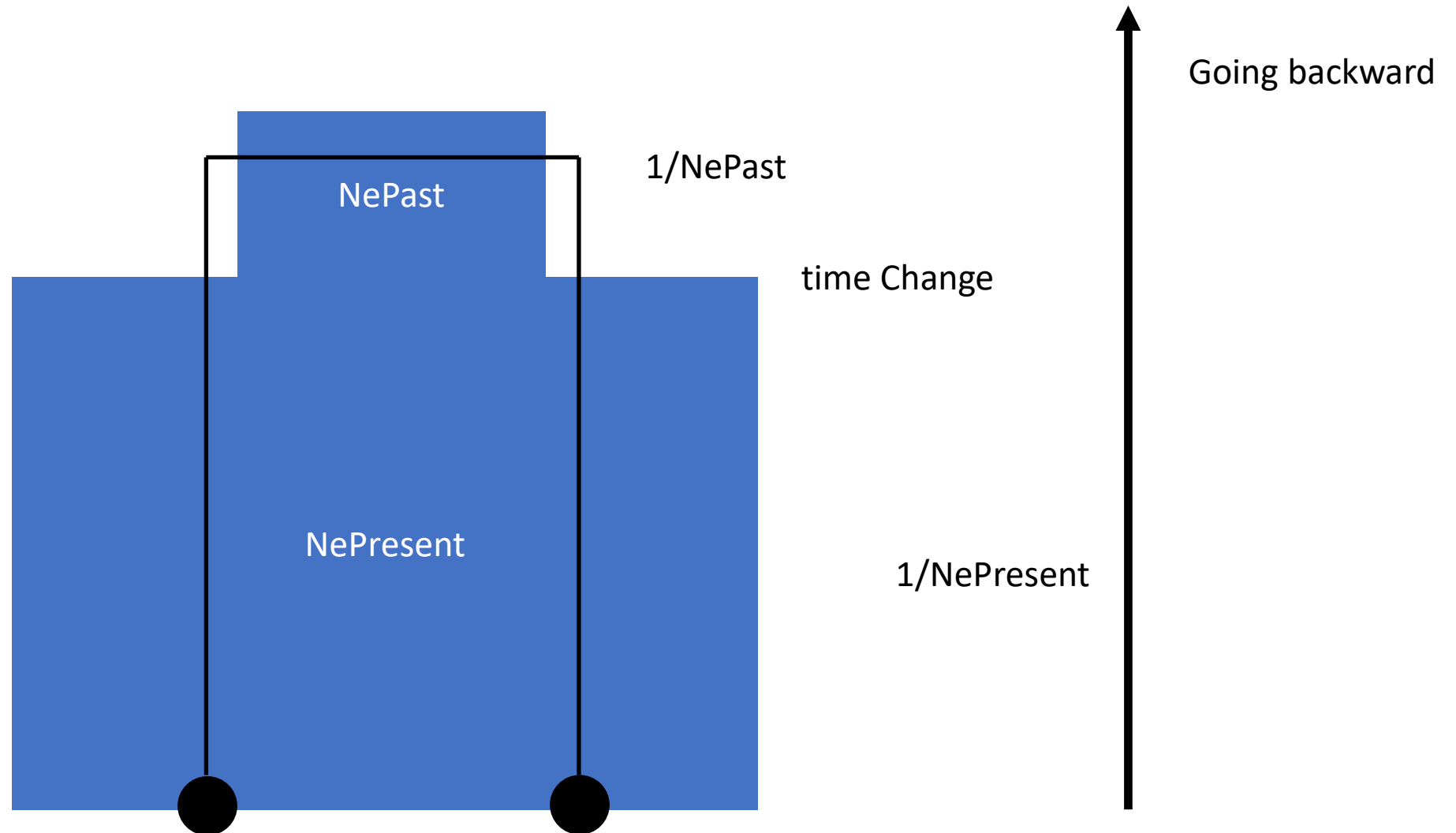
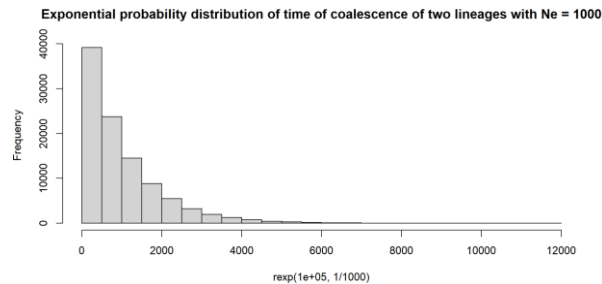
Model Ne changes

Coalescence follows
Exponential with
parameter $1/\text{NePast}$
conditioned it did
not reach
coalescence in the
previous Ne



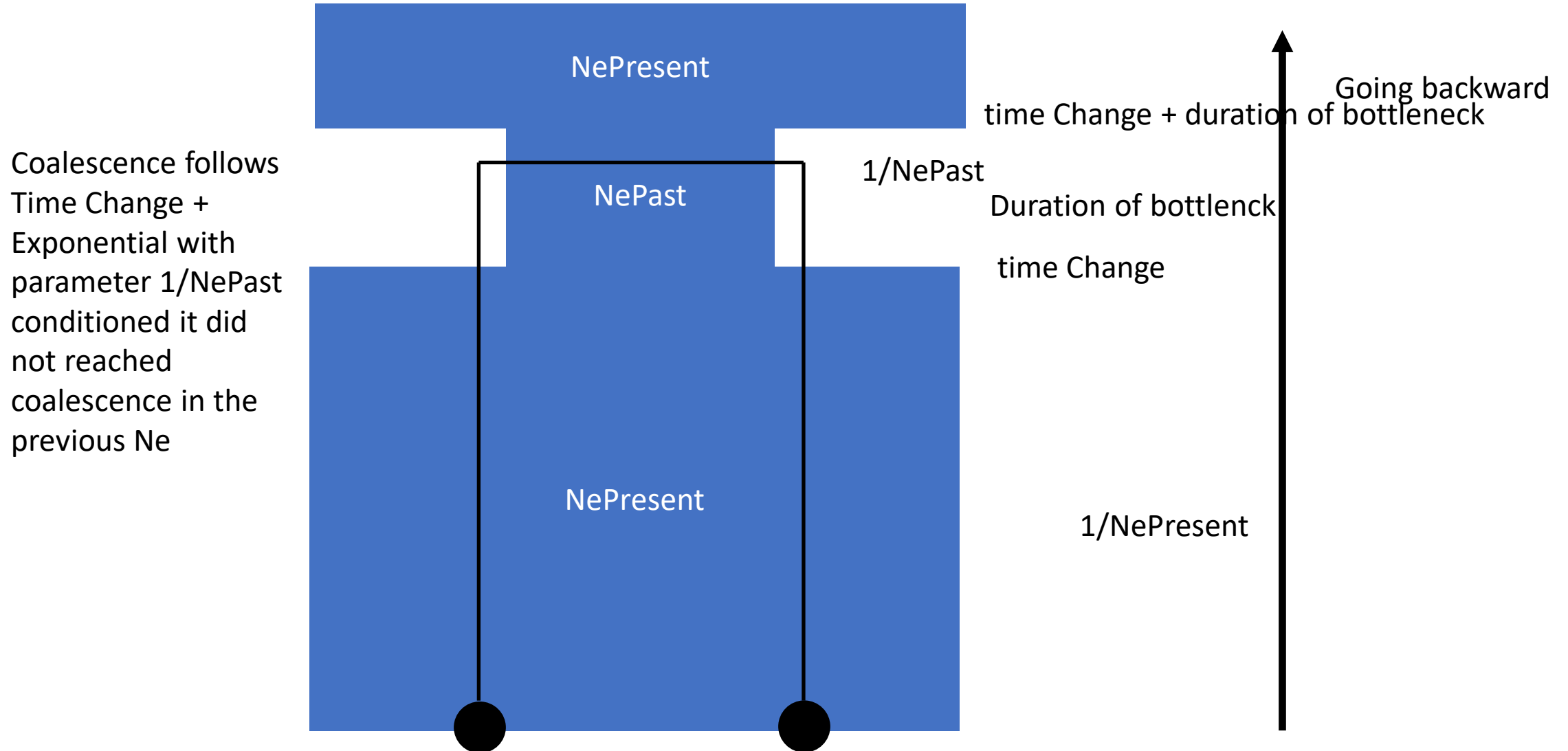
Model Ne changes

Coalescence follows
time Change +
Exponential with
parameter $1/\text{NePast}$
conditioned it did
not reached
coalescence in the
previous Ne



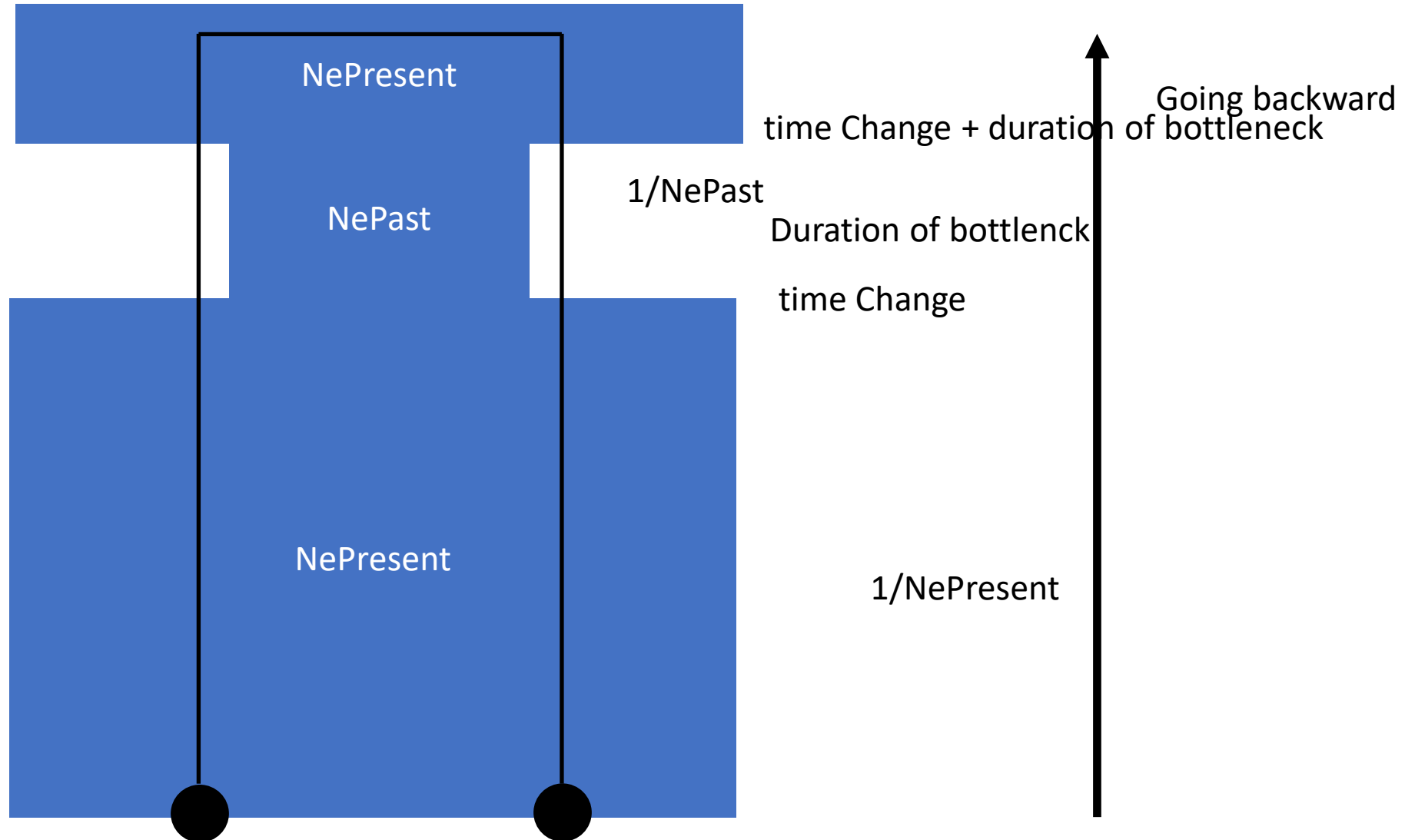
Pseudocode?

Model Ne bottleneck

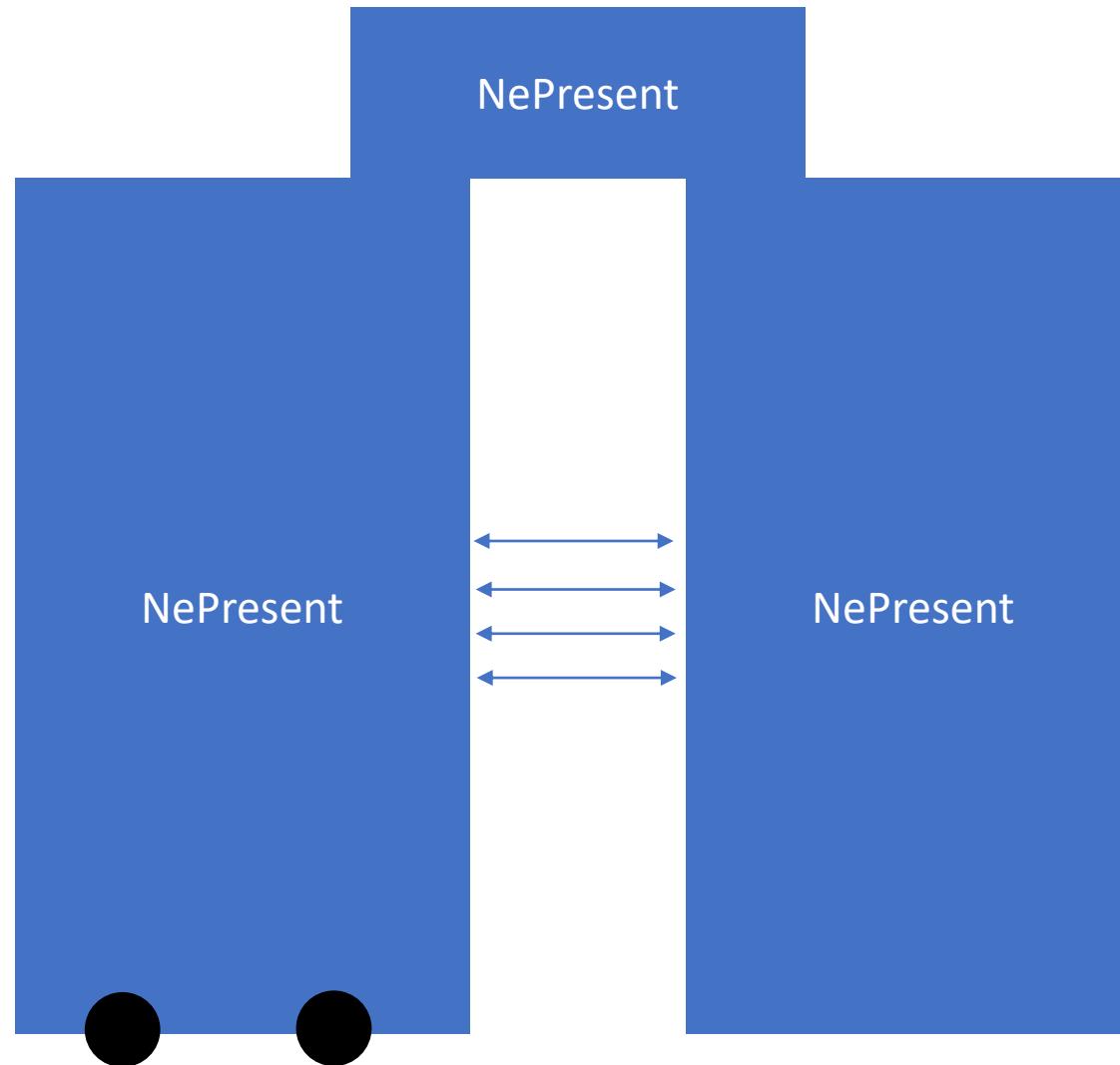


Model Ne bottleneck

Coalescence follows
Time Change
+duration of
bottleneck +
Exponential with
parameter
 $1/\text{NePresent}$
conditioned it did
not reached
coalescence in the
previous Ne and
NePast



Model Ne Structured



RELATE

- ◇ Step 0: Install Relate

```
git clone https://github.com/relate/Relate.git
cd Relate
make
```

- ◇ Step 1: Prepare Input Files

You need:

phased.vcf.gz: Phased and filtered VCF file

genetic_map.txt: Genetic map in 3-column format: chr pos cM

 Optional:

mask.bed: BED file to exclude low-quality or recombination-poor regions

Convert VCF to .haps and .sample:

```
RelateFileFormats/ConvertFromVcf \
```

```
--input phased.vcf.gz \
```

```
--output relate_input \
```

```
--sample sample_list.txt \
```

```
--haps-format \
```

```
--map genetic_map.txt
```

RELATE

- ◇ Step 2: Run Relate (tree building + mutation inference)

*Relate *

```
--mode All \  
--haps relate_input.haps \  
--sample relate_input.sample \  
--map genetic_map.txt \  
--output relate_output \  
--threads 8
```

This will generate:

.anc = Ancestral recombination graph (trees)

.mut = Mutation placements

- ◇ Step 3: Estimate Population Size History

*EstimatePopulationSize/EstimatePopulationSize.sh *

```
-i relate_output \  
-m 1.25e-8 \  
-o popsize_output
```

Parameters:

-m: Per-base mutation rate (e.g., 1.25e-8 for humans)

-o: Output prefix

This produces:

popsize_output.coal = Coalescent rate summaries

popsize_output.Ne = Effective population size over time

Plotting scripts use this directly

RELATE

◇ Step 4: Plot Population Size (Optional)

Visualize the Ne over time from popsize_output.Ne.

Simple example (in R):

```
popsiz <- read.table("popsiz_output.Ne", header=TRUE)
plot(popsiz$Time, popsiz$Ne, type='l', log='x',
     xlab="Generations ago", ylab="Effective population size",
     main="Population size history")
```

Input File Summary

File	Description
phased.vcf.gz	Phased SNP data
genetic_map.txt	Recombination map (chr, pos, cM)
sample_list.txt	(Optional) list of samples to include
mask.bed	(Optional) to exclude problematic regions

Relate assumes constant mutation rate and good phasing — use SHAPEIT4 or Eagle for preprocessing.

Works best on large sample sizes (e.g., >20 haploid genomes) for demographic signal.

Estimates are relative to mutation rate — errors in -m affect Ne scale.